

# Richard van Zyl — Skills Matrix (2026)

Professional years of experience, a SQL performance deep-dive, and a project-by-project breakdown per employer.

**Years column.** Professional years only, counted from paid work starting **April 2017** — nothing exceeds **9**. Formal study (Varsity College, 2014–2016) is not counted. Engagement-tied skills (PostgreSQL, Redis, MongoDB, Azure Functions) use that engagement's span, not the whole career.

| Section                   | What's in it                                       |
|---------------------------|----------------------------------------------------|
| Skills by area            | Grouped tables, strongest first within each group  |
| SQL performance deep-dive | Execution plans, indexing, locking, scale patterns |
| Employment & projects     | Employer metadata, duties, and project cards       |

## Skills by area

Within each table, rows are ordered by years (highest first).

### Languages

| Skill             | Years |
|-------------------|-------|
| C# / .NET         | 9     |
| T-SQL             | 9     |
| PostgreSQL        | 7     |
| TypeScript        | 5     |
| JavaScript        | 5     |
| KQL               | 5     |
| Batch Script      | 5     |
| Visual Basic .NET | 3     |
| Java              | 2     |

### Data formats

| Skill | Years |
|-------|-------|
| JSON  | 9     |
| XML   | 8     |
| XAML  | 7     |
| YAML  | 6     |
| HTML  | 3     |

### Databases & data engineering

| Skill                          | Years |
|--------------------------------|-------|
| SQL Server / SSMS              | 9     |
| SQL performance tuning         | 8     |
| Azure SQL                      | 7     |
| MySQL Workbench                | 6     |
| Azure Data Studio              | 6     |
| Redis (distributed caching)    | 5     |
| SSRS                           | 5     |
| MongoDB                        | 4     |
| Table partitioning & archiving | 4     |
| In-Memory OLTP                 | 4     |

## Architecture & design

| Skill                                 | Years |
|---------------------------------------|-------|
| Clean Architecture                    | 9     |
| Domain-Driven Design (DDD)            | 8     |
| Microservices                         | 7     |
| Multi-tenancy / tenant isolation      | 7     |
| CQRS                                  | 6     |
| Event-driven architecture             | 5     |
| Idempotency / exactly-once processing | 4     |

## Business analysis & solution design

| Skill                                              | Years |
|----------------------------------------------------|-------|
| Business & problem-domain analysis                 | 9     |
| Requirements specification (BRS / URS / FRS / TRS) | 9     |
| Solution design & solution architecture            | 9     |
| Technical documentation (audience-tailored)        | 9     |
| Stakeholder engagement & translation               | 8     |
| Process & system modelling (UML)                   | 8     |

## Backend frameworks & libraries

| Skill                   | Years |
|-------------------------|-------|
| Entity Framework        | 9     |
| MediatR                 | 7     |
| AutoMapper              | 6     |
| FluentValidation        | 6     |
| ASP.NET                 | 3     |
| Circuit breaker (Polly) | 3     |
| SignalR / WebSockets    | 3     |

## Frontend & UI

| Skill            | Years |
|------------------|-------|
| Angular          | 6     |
| Angular Material | 6     |
| Ajax             | 6     |
| MVC              | 5     |
| Node.js          | 5     |

## Delivery practices

| Skill                                   | Years |
|-----------------------------------------|-------|
| Scrum                                   | 8     |
| Unit testing (xUnit / NUnit / Moq)      | 6     |
| OAuth2 / OpenID Connect / JWT           | 6     |
| Integration testing (TestContainers)    | 5     |
| Feature flags (Azure App Configuration) | 4     |

## Cloud, DevOps & tooling

| Skill         | Years |
|---------------|-------|
| Visual Studio | 9     |

| Skill                                         | Years |
|-----------------------------------------------|-------|
| VS Code                                       | 9     |
| GitHub                                        | 9     |
| Azure DevOps (previously TFS)                 | 9     |
| Postman                                       | 9     |
| Swagger                                       | 9     |
| Enterprise Architect (UML)                    | 8     |
| Azure CI/CD                                   | 7     |
| Docker                                        | 7     |
| Pipeline-as-code (Azure DevOps YAML)          | 6     |
| Message queues (Azure Service Bus / RabbitMQ) | 6     |
| Telerik                                       | 6     |
| Azure Monitor / Log Analytics / App Insights  | 5     |
| Microsoft Azure Insights                      | 5     |
| New Relic (APM, release tracking, alerting)   | 4     |
| API Gateway (Azure API Management)            | 4     |
| Elastic Stack (ELK) / Serilog                 | 4     |
| OpenTelemetry                                 | 3     |
| Grafana                                       | 3     |
| Graylog                                       | 3     |
| DevExpress                                    | 3     |

Security & compliance

| Skill                                       | Years |
|---------------------------------------------|-------|
| Role-based access control (RBAC)            | 8     |
| Data-protection-aware design (POPIA / GDPR) | 5     |
| SOX-aligned data segregation                | 4     |

Deep dive: SQL performance tuning

8 years · High-volume financial ledger systems integrating 150+ money processors across African markets.

| Sub-skill               | Focus                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------|
| Execution plan analysis | SHOWPLAN, SET STATISTICS IO / TIME; spotting scans, key lookups, and high-cost operators                   |
| Indexing strategy       | Clustered, non-clustered, filtered, and covering indexes; fragmentation and rebuild / reorganise decisions |
| Locking & concurrency   | Isolation levels (READ COMMITTED → SNAPSHOT); deadlock detection via Profiler / Extended Events            |
| Tuning patterns         | Correlated subqueries → joins; INNER JOIN vs EXISTS; SARGable predicates; avoiding scalar UDFs             |
| Design for scale        | Date/tenant partitioning; indexed / materialised views; high-volume batch inserts (ledger entries)         |
| In-Memory OLTP          | Memory-optimised tables and natively compiled procedures on high-contention paths                          |
| Query Store             | Regression tracking, forced parameterization, plan forcing                                                 |
| Health monitoring       | sys.dm_os_wait_stats, sys.dm_db_index_usage_stats, sp_WhoIsActive                                          |

## Employment & projects

Each block uses the same shape: metadata → duties → projects (role line, summary, tech). Side projects are labelled as such — they are not current employment.

### Yuno Technologies (*side project — not employment*)

| Field      | Detail                                                                   |
|------------|--------------------------------------------------------------------------|
| Industry   | Digital consultancy                                                      |
| Role       | Stakeholder / Solutions Architect                                        |
| Period     | Apr 2026 – present                                                       |
| Operating  | Partner-run; Richard contributes architecture, not day-to-day operations |
| Consulting | Yes (client names held private)                                          |

**Duties** - Stakeholder contributing architecture, backend engineering, and security design across client platforms. - **Current production:** four-plane deployment topology with one-way initiation; BFF holding session rather than tokens; response-only identity plane on Keycloak; containerised platform across environments. - **Phase 2 / planned:** inter-plane trust via mTLS / SPIFFE/SPIRE behind a documented interim. - **Under evaluation:** ledger-plane isolation via KVM-backed microVM separation so a container escape in an adjacent workload cannot reach the ledger. - ADRs with formal supersession across topology changes.

**Tech:** .NET · Vite / React · PostgreSQL · Redis · RabbitMQ · gRPC · Django / DRF · Azure Functions · Blob Storage · Keycloak · Prometheus / Grafana / ELK

### Orchestration Platform (*side project — not employment*)

| Field     | Detail                                                           |
|-----------|------------------------------------------------------------------|
| Industry  | Business orchestration                                           |
| Role      | Co-founder                                                       |
| Period    | current                                                          |
| Operating | Side project; public at the same altitude as the site “Now” card |

**Duties** - Deterministic business-orchestration platform on gRPC and standalone module assemblies; deterministic replay as the differentiator. - Owns the core engine, runtime, module execution, configuration and flow state, traces and audit logs.

**Tech:** gRPC · standalone module assemblies

### Raging River Trading (Pty) Ltd

| Field      | Detail                                                                                                     |
|------------|------------------------------------------------------------------------------------------------------------|
| Industry   | Fintech / Gaming                                                                                           |
| Role       | Solutions Architect / Technical Lead — Core Financial Systems ( <i>payroll title: Software Developer</i> ) |
| Period     | Dec 2022 – Mar 2026                                                                                        |
| Consulting | No                                                                                                         |

**Duties** - Led the rewrite of a legacy banking platform into a white-label financial engine for deposits & withdrawals across African markets (including Betway and Jackpot City). - Engineered for horizontal and vertical scale under concurrent ledger load. - Integration platform for 150+ money processors — IronPython marshalling in .NET; rewrote per-integration isolation to stop shared thread-pool / socket exhaustion. - Migrated ~150 million records on a peak-trading Saturday with no New Relic Apex drop.

#### Project — Core Financial Systems

##### Solutions Architect / Technical Lead · Team 1–4 · Backend

Full-scale rewrite to a white-label engine handling deposits & withdrawals for multiple brands. Horizontal scale with APIs that safely touch a concurrent ledger. Schema-per-brand under one SQL Server (separate databases priced out); transactional outbox to per-brand queues with dead-lettering.

**Tech:** Azure SQL · C# / .NET · Docker · RabbitMQ · EF Core

#### Project — Integration layer & back-office APIs

##### Solutions Architect / Technical Lead · Team 5 · Backend

Integration layer abstracting 150+ money processors. Back-office APIs for safe concurrent ledger access under high volume.

**Tech:** C# / .NET · Azure SQL

## Project — High-availability & concurrency design

### Solutions Architect / Technical Lead · Team 1 · Backend

HA architecture for concurrent financial transactions — idempotent attempt records with rowversion optimistic concurrency on the ledger.

**Tech:** C# / .NET · Azure SQL

## MeterMo

| Field      | Detail                                                                |
|------------|-----------------------------------------------------------------------|
| Industry   | Utilities — automated electricity, water and gas metering & reporting |
| Role       | Software Developer                                                    |
| Period     | Apr 2022 – Dec 2022                                                   |
| Consulting | No                                                                    |
| Leaving    | Mandate complete; sought a larger engineering team                    |

### Duties

Platform modernisation across the metering estate. Migrated six projects from TFS to Azure DevOps and upgraded frameworks/packages; maintained two APIs, two cross-platform field apps (Xamarin, Cordova), and two websites.

**Tech:** C# / .NET · ASP.NET · Angular · Xamarin · Cordova · Azure DevOps · TFS

## Dotcom Software Solutions

| Field      | Detail                                         |
|------------|------------------------------------------------|
| Industry   | Consulting (Fintech)                           |
| Role       | Software Developer (Specialist Problem-Solver) |
| Period     | Jan 2022 – Apr 2022                            |
| Consulting | Yes                                            |
| Leaving    | No longer remote / not a culture fit           |

### Duties

Specialist problem-solver for challenges others could not resolve — full-stack fintech delivery for Nedbank, Standard Bank, and PSG Wealth. Business analysis including Schemes Bot monetisation design and payment validations.

## Project — Schemes Bot (Standard Bank)

### Software Developer · Team 5 · Backend · Jan–Feb 2022

BotFramework backend upgrade **to .NET 6 from .NET Standard**, including dependency updates and breaking-change fixes. Prior attempts had been rolled back; completed and live in production.

**Tech:** Azure SQL · BotFramework · C# / .NET 6 · MediatR · QnA Maker (*historical / retired*)

## Project — PSG Wealth website

### Software Developer / Analyst / Architect · Team 6 · Backend · Feb–Mar 2022

Angular Universal SSR for SEO — unblocked a two-year struggle in one week.

**Tech:** Docker · Angular · Bootstrap · Material · Node.js · C# / .NET 6 · Umbraco CMS

## Payteq

| Field      | Detail                                          |
|------------|-------------------------------------------------|
| Industry   | Fintech                                         |
| Role       | Senior Developer / Business Analyst / Team Lead |
| Period     | Apr 2020 – Jan 2022                             |
| Consulting | No                                              |
| Leaving    | Career growth                                   |

### Duties

Full-stack work on EDI, GoTrips, and Veriseal; business analysis with the BA team; team lead on the GoTrips rewrite, optimisation, and architecture.

### Project — GoTrips

**Software Developer / Analyst / Team Lead · Team 5 · Full stack · Jan 2021 – Jan 2022**

Ground-up rewrite eliminating accumulated technical debt; coding standards, design principles, performance work, and automation for a maintainable system.

**Tech:** C# / .NET Core API · PostgreSQL · Azure CI/CD · FluentValidation · Angular · Azure Functions

### Project — Veriseal

**Software Developer / Analyst / Architect · Team 1 · Full stack · side project, migrated opportunistically**

Independently migrated a critical compliance platform (KYC, AML, mass payments, bank-account verification) from legacy ASP.NET to .NET Core API with Angular 8.

**Tech:** C# / .NET Core · SQL · SSRS · Angular · ASP.NET · .NET Framework · FluentValidation · AutoMapper

### Project — GoBills

**Team Lead · Team 2 · Full stack · Apr 2020 – Jan 2021**

Business-management solution (inventory, POS, financials) with sophisticated RBAC for permissions and client access.

**Tech:** C# / .NET Core API · Identity Framework · DI · custom CMS · Angular 8 · Bootstrap · Angular Material

### Project — Interchange IDE & Insight Server

**Development & maintenance · Team 2 · Support**

Messaging-middleware integration platform with RBAC; ongoing SaaS support and maintenance.

**Tech:** SQL · C# / .NET Core · IP microservices / integrations

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## iPlan Global

| Field      | Detail                                                           |
|------------|------------------------------------------------------------------|
| Industry   | Supply chain / manufacturing / farming / industrial / automation |
| Role       | Software Developer / Business Analyst                            |
| Period     | Oct 2017 – Apr 2020                                              |
| Consulting | Yes                                                              |
| Leaving    | Career growth                                                    |

### Duties

Team and solo delivery; productivity automation; BA work across mining, industrial, supply chain, manufacturing, and sales — needs analysis, domain analysis, solution design and architecture.

### Project — Consolidated Job Dashboard

**Software Developer / Analyst / Architect · Team 1 · Full stack · May–Jun 2019**

Supply-chain consolidation of job material requirements — replaced a manual, error-prone process.

**Tech:** VBScript · SQL

### Project — Espresso Quote Application

**Software Developer / Analyst / Architect · Team 1 · Full stack · Feb–Apr 2019**

Multi-platform quoting for sales reps — on-the-spot quotes, email delivery, manager approval.

**Tech:** jQuery · SQL · C# / .NET · JavaScript · Entity Framework · MVC · Ajax

### Project — ASP.NET Food Portal

**Development & maintenance · Team 2 · Full stack · May–Sep 2019**

Farmers log deliveries and poison-testing results with multi-location security and compliance.

**Tech:** ASP.NET · C# / .NET · XAML · MVVM

### Project — Espresso RepCheckIn

**Software Developer / Architect · Team 1 · Full stack · May–Sep 2019**

Rep tracking and tiered priority visit lists for optimised field routes.

**Tech:** JavaScript · C# / .NET · jQuery · MVC · Ajax

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## Novigo (Pty) Ltd

| Field      | Detail                           |
|------------|----------------------------------|
| Industry   | Recruitment                      |
| Role       | Software Developer (C# / .NET)   |
| Period     | Apr 2017 – Oct 2017              |
| Consulting | Yes (the company was the client) |
| Leaving    | Contract ended                   |

### Duties

Business and problem-domain analysis; solution planning; framework design; system and server implementation; ERP maintenance; Linux server and domain/hosting support.

### Project — Novigo Management System

**Sole developer · Team 1 · Full stack · Apr–Sep 2017**

Designed, built, and deployed a complete ERP from scratch, including server setup and IT support.

**Tech:** MySQL · C# / .NET · Linux · WinForms